

# Tableau Prep + Python Integration Using TabPy

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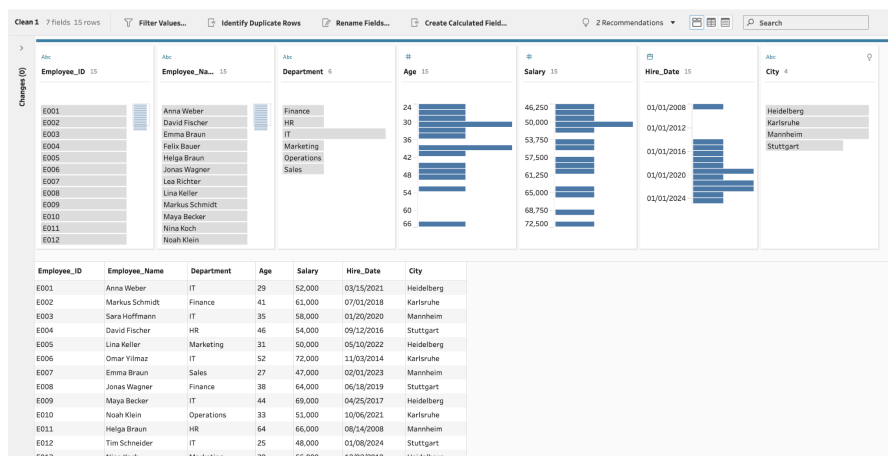
## Dataset Used

- File: employee\_tableau\_prep\_python\_sample.csv
- Data includes:
  - Employee ID
  - Employee Name
  - Department
  - Age
  - Salary

## Workflow Overview

1. Connected Tableau Prep to the employee CSV dataset.
2. Added a Script Step in Tableau Prep.
3. Connected the Script Step to the TabPy server.
4. Executed a Python script using Pandas.
5. Filtered employees belonging to the IT department.
6. Calculated the average salary of all employees.
7. Identified the oldest employee in the organization.
8. Returned transformed results back into Tableau Prep.

## Screenshots



The screenshot displays the Tableau Prep interface. At the top, there's a toolbar with options like 'Filter Values...', 'Identify Duplicate Rows', 'Rename Fields...', and 'Create Calculated Field...'. Below this, a data table is shown with columns: Employee\_ID, Employee\_Name, Department, Age, Salary, Hire\_Date, and City. The table contains 13 rows of employee data. To the right of the table, a visualization is shown, which is a horizontal bar chart. The chart has 'Employee\_ID' on the x-axis and 'Salary' on the y-axis. The bars represent the salary for each employee, with the highest salary being 72,500 (Employee E011) and the lowest being 46,250 (Employee E001).

| Employee_ID | Employee_Name  | Department | Age | Salary | Hire_Date  | City       |
|-------------|----------------|------------|-----|--------|------------|------------|
| E001        | Anna Weber     | IT         | 29  | 52,000 | 03/15/2021 | Heidelberg |
| E002        | Markus Schmidt | Finance    | 41  | 61,000 | 07/01/2018 | Karlsruhe  |
| E003        | Sara Hoffmann  | IT         | 35  | 58,000 | 01/20/2020 | Mannheim   |
| E004        | David Fischer  | HR         | 46  | 54,000 | 09/12/2016 | Stuttgart  |
| E005        | Lina Keller    | Marketing  | 31  | 50,000 | 05/10/2022 | Heidelberg |
| E006        | Onar Yilmaz    | IT         | 52  | 72,000 | 11/03/2014 | Karlsruhe  |
| E007        | Emma Braun     | Sales      | 27  | 47,000 | 00/01/2023 | Mannheim   |
| E008        | Jonas Wagner   | Finance    | 38  | 64,000 | 06/18/2019 | Stuttgart  |
| E009        | Maya Becker    | IT         | 44  | 69,000 | 04/25/2017 | Heidelberg |
| E010        | Noah Klein     | Operations | 33  | 51,000 | 10/06/2021 | Karlsruhe  |
| E011        | Helga Braun    | HR         | 64  | 66,000 | 08/14/2008 | Mannheim   |
| E012        | Tim Schneider  | IT         | 25  | 48,000 | 01/08/2024 | Stuttgart  |
| E013        | Nina Koch      | Marketing  | 39  | 56,000 | 12/02/2019 | Heidelberg |

Diagram showing the data flow: employee\_tabl... → Clean 1 → Script 1 → Output.

Output window showing 8 fields:

| Oldest_Employee_Department | Oldest_Employee_Name | Average_Salary_All_Employees | Employee_ID | Employee_Name | Department | Age | Salary |
|----------------------------|----------------------|------------------------------|-------------|---------------|------------|-----|--------|
| HR                         | Helga Braun          | 57,100                       | E001        | Anna Weber    | IT         | 29  | 52,000 |
| HR                         | Helga Braun          | 57,100                       | E003        | Sara Hoffmann | IT         | 35  | 58,000 |
| HR                         | Helga Braun          | 57,100                       | E006        | Omar Yilmaz   | IT         | 52  | 72,000 |
| HR                         | Helga Braun          | 57,100                       | E009        | Maya Becker   | IT         | 44  | 69,000 |
| HR                         | Helga Braun          | 57,100                       | E012        | Tim Schneider | IT         | 25  | 48,000 |

Save to Output.hyper

| Oldest_Employee_Department | Oldest_Employee_Name | Average_Salary_All_Employees | Employee_ID | Employee_Name | Department | Age | Salary |
|----------------------------|----------------------|------------------------------|-------------|---------------|------------|-----|--------|
| HR                         | Helga Braun          | 57,100                       | E001        | Anna Weber    | IT         | 29  | 52,000 |
| HR                         | Helga Braun          | 57,100                       | E003        | Sara Hoffmann | IT         | 35  | 58,000 |
| HR                         | Helga Braun          | 57,100                       | E006        | Omar Yilmaz   | IT         | 52  | 72,000 |
| HR                         | Helga Braun          | 57,100                       | E009        | Maya Becker   | IT         | 44  | 69,000 |
| HR                         | Helga Braun          | 57,100                       | E012        | Tim Schneider | IT         | 25  | 48,000 |

Script 1 8 fields 5 rows Filter Values...

Settings

Connection type

☐ Rserve

☒ Tableau Python (TabPy) Server

Server

Connection to localhost: 9004

File Name

main 1.py

Browse

Function Name

run\_script

Include the schema function "get\_output\_schema" in your script to define the fields and data types that are returned. [Learn more](#)

## Final Output

The final output contains:

- Only IT department employees.
- Average salary information for all employees.
- Oldest employee name.
- Oldest employee department.